Search for Time-Reversal-Symmetry Violation in Thallium Fluoride Using a Cryogenic Buffer-Gas Beam Source

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Outline

1. Purpose of the experiment
2. Experimental apparatus
3. Beam source
4. Magnetic shield
5. Other topics
6. Conclusion

Purpose of the experiment

- Motivation
- T-violation and EDMs
- Schiff theorem
- Schiff moment
- Molecules and TIF
- Separated oscillatory coils

Motivation (1)

Baryon asymmetry of the universe

- Number density of antibaryons is almost negligible:

$$\eta = \frac{n_{\mathrm{B}} - n_{\bar{\mathrm{B}}}}{n_\gamma} \approx \frac{n_{\mathrm{B}}}{n_\gamma} \approx 10^{-10}$$

- Standard model / GR does not explain why that should be
- Sakharov conditions (1967): require *CP*-symmetry violation
  - By CPT symmetry, this implies *T*-symmetry violation
  - Standard model does not include enough *CP* (or *T*) violation
  - Need to look for *T*-violating phenomena

Motivation (2)

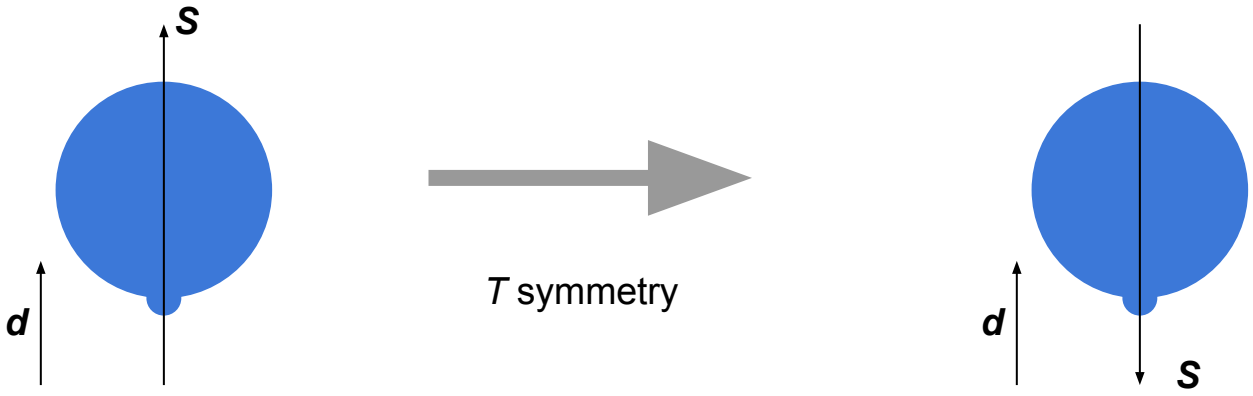
Strong CP problem

- 1960s experiments showed *CP* is broken in the weak interaction
- *CP*-violation never observed in strong interactions
- "Amount" of *CP*-violation related to neutron EDM:

  $d_N \approx \theta \cdot (10^{-16}\ \text{e cm})$ [2, eqn 29.111]
- Latest measurements constrain strong *CP*-violation to a $\theta < 10^{-10}$ effect
- Better EDM measurements can further constrain θ

T-violation and EDMs

EDMs in fundamental particles violate *T* symmetry:



Particle with EDM parallel to spin

“Spins” the other way, but EDM still points up

Schiff theorem

- Want to look for nuclear EDMs in atoms/molecules
- But that's (almost) impossible:

  Schiff theorem: EDM of a non-relativistic point particle in a neutral atom or molecule causes no 1st-order energy shifts

- Ways around the Schiff limitation:
  - Relativistic effects
  - Finite nuclear size

Schiff moment

- For finite-size nucleus in a neutral atom in external electric field, there *is* an observable (1st-order) energy shift
- Shift is proportional to the Schiff moment

$$\mathbf{S} = \frac{Ze}{10}\left[\langle R^2\mathbf{R}\rangle - \frac{5}{3}\langle R^2\rangle\langle\mathbf{R}\rangle\right]$$

- The effect is that of a off-diagonal interaction

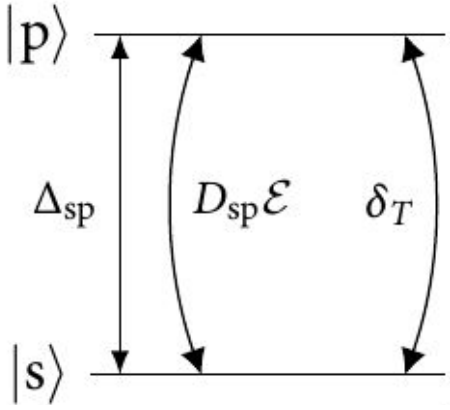
$$\langle \mathrm{s}|\delta H|\mathrm{p}\rangle = \langle \mathrm{s}| - e\phi(\mathbf{r})|\mathrm{p}\rangle \qquad \text{with} \quad \phi(\mathbf{r}) \equiv -4\pi\,\mathbf{S}\cdot\nabla\delta^3(\mathbf{r})$$

- Schiff moment can be related to measured energy shifts

$$S = \frac{\nu}{1.98\cdot 10^6}\frac{e\ \mathrm{fm}^3}{\mathrm{Hz}}$$

[3, eqn 22]

Polarization



Consider an off-diagonal interaction *H* in a two-level atom.

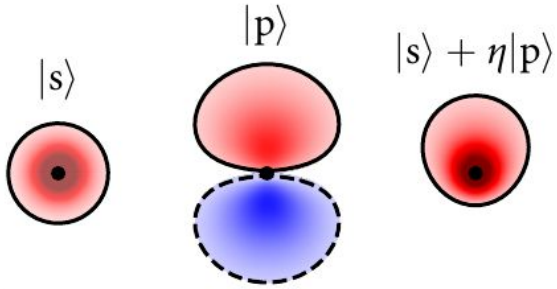
In a general superposition state

$$|\psi\rangle \equiv |\mathrm{s}\rangle + \eta|\mathrm{p}\rangle$$

its expectation value is

$$\langle \delta H\rangle \equiv \langle \psi|\delta H|\psi\rangle = 2\eta\langle \mathrm{s}|\delta H|\mathrm{p}\rangle;$$

i.e., the states have to be mixed (η ≠ 0) for a first-order energy shift.



Why molecules?

Atoms: Energy spacing ~ eV, so difficult to polarize.

Molecules: Closely-spaced levels available, easy to polarize.

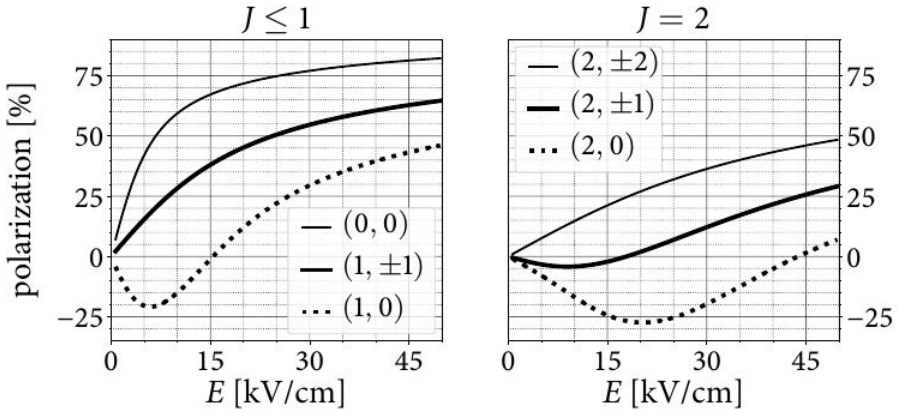
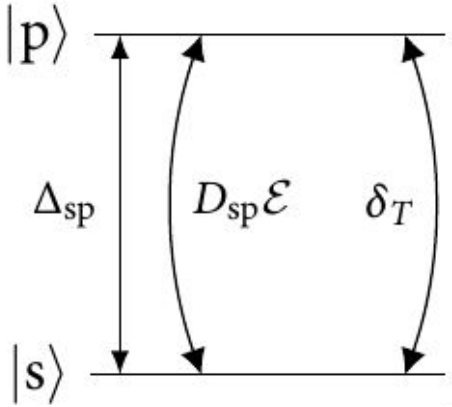
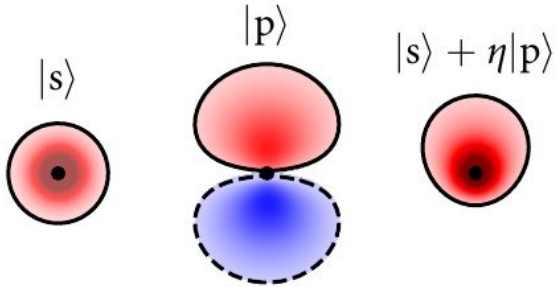


Figure 3: Degree of polarization of TlF molecules in an external field.





Why TIF?

Recall the matrix element of the interaction is

$$\langle \mathrm{s}|\delta H|\mathrm{p}\rangle = 4\pi e\,\mathbf{S}\cdot\nabla(\psi_{\mathrm{s}}^{*}\psi_{\mathrm{p}})_{r\to 0}$$

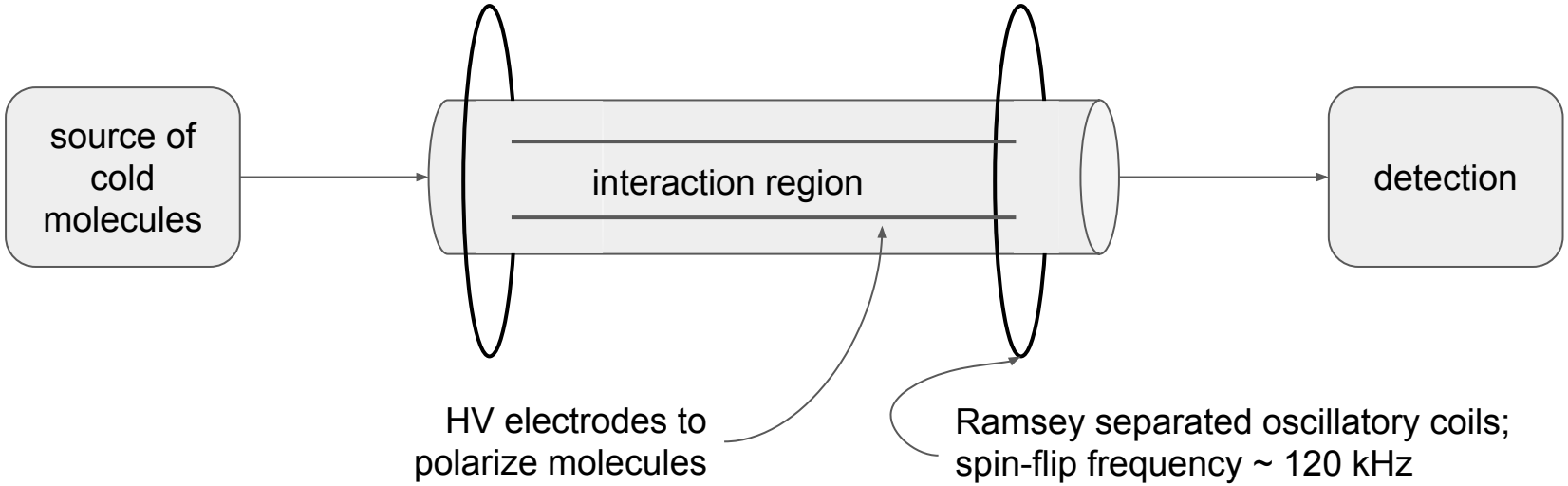
where, approximately,

$$\nabla(\psi_{\mathrm{s}}^{*}\psi_{\mathrm{p}}) = \frac{Z^2}{a_0^4} \qquad |\mathbf{S}| = d_{\mathrm{N}}R_0^2/10$$

So want heavy & large nuclei, like thallium (Z = 81, A = 205).

<u>Also:</u> optical cycling transition in TIF for detection & cooling.

Energy shift measurement



- Purpose of the experiment
- Experimental apparatus
- Beam source
- Magnetic shield
- Other topics
- Conclusion

Experimental apparatus

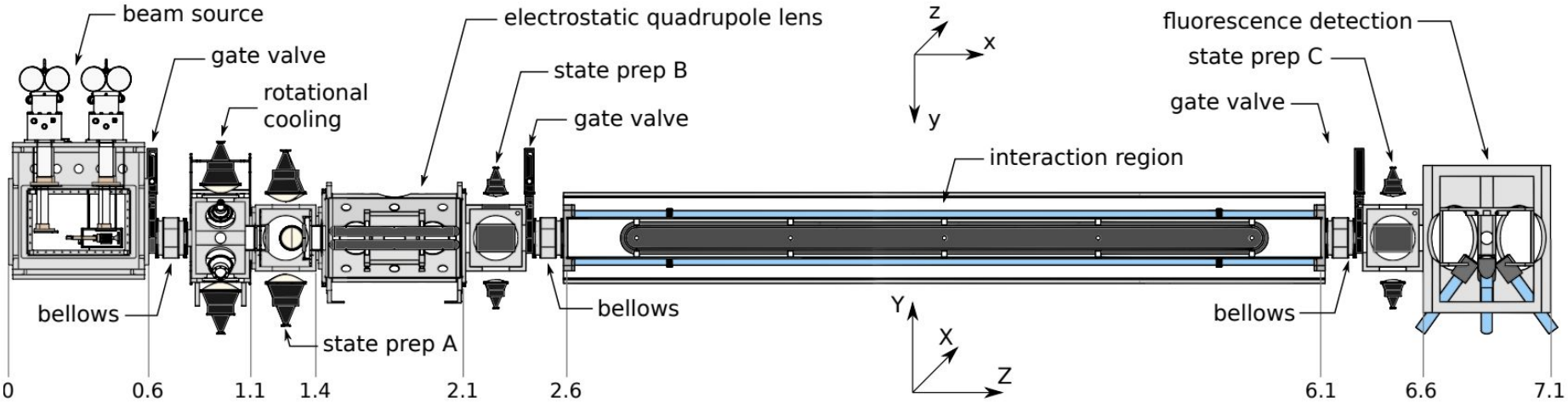


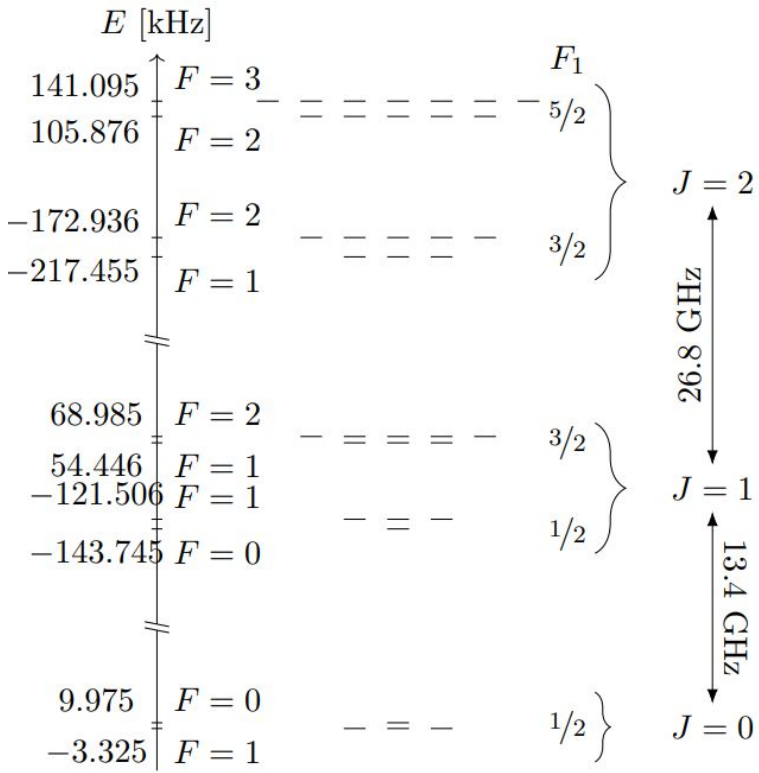
Fig. ref.: **[1]**

TIF ground state energy levels

We make use of several levels of the molecule:

- Beam source thermally populates all levels
- Lens focuses $|J = 2, m_J = 0\rangle$
- Interaction region uses a superposition of two $J = 1$ states

State preparation regions are used to transfer thermal population into the “useful” states.



Overview of the experiment (1 / 2)

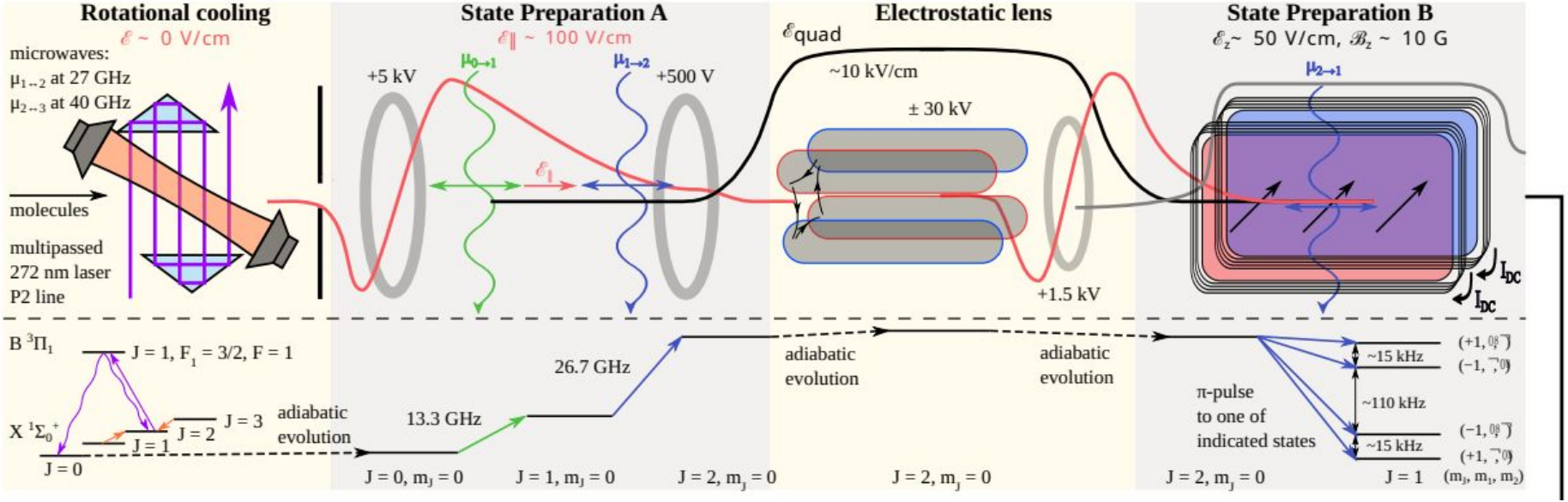


Fig. 6 of ref. **[1]**

Overview of the experiment (2 / 2)

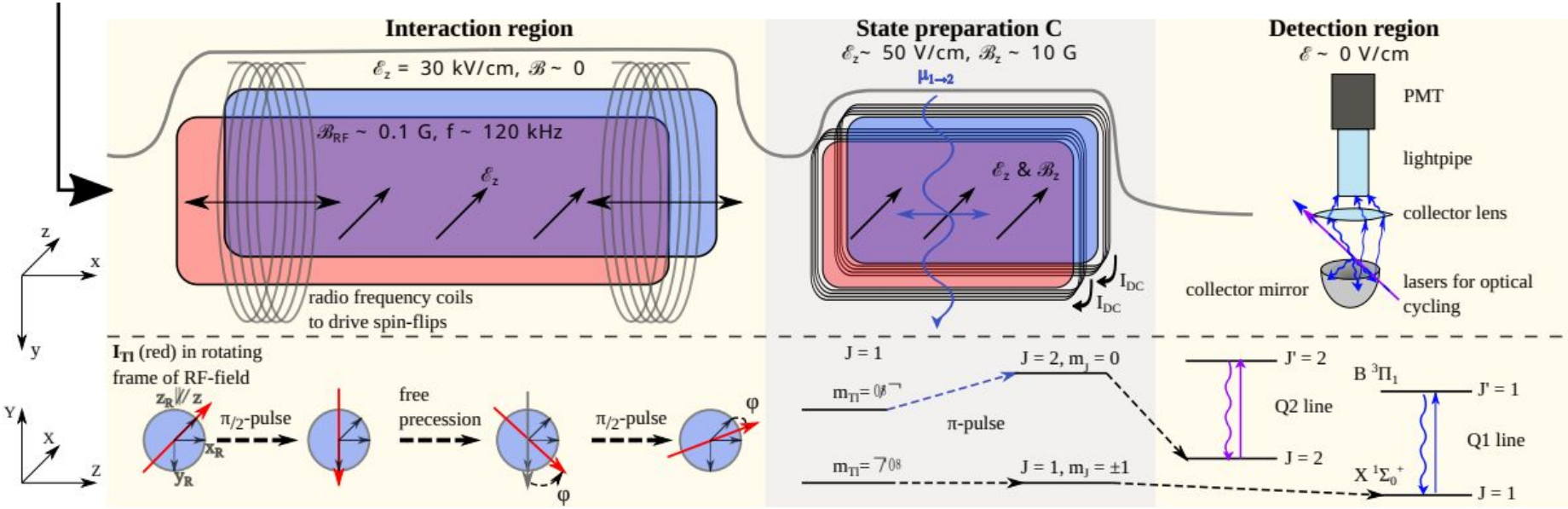
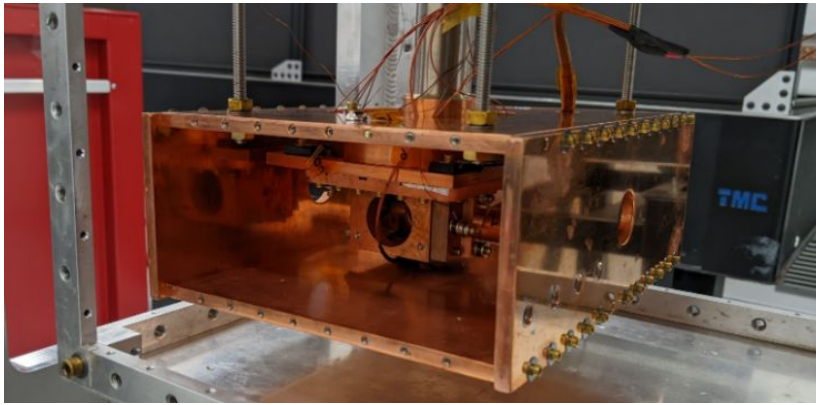


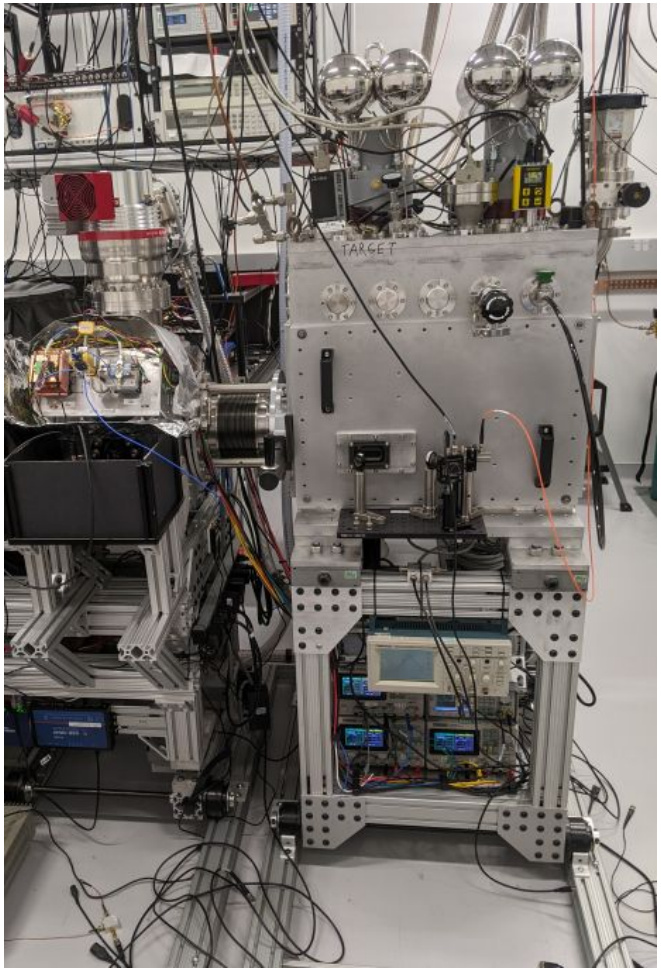
Fig. 6 of ref. **[1]**

Beam source

Purpose: produce a cold beam of TlF

Method: cryogenic buffer-gas laser ablation



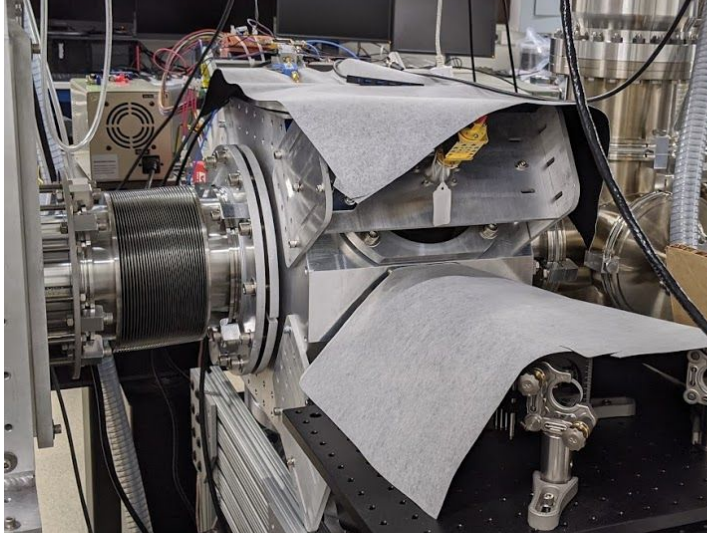


To be discussed in more detail shortly …

Rotational cooling

<u>Purpose:</u> concentrate population from $J = 1, 2, 3$ into one hyperfine state in $J = 0$

<u>Method:</u> optical pumping, microwave-driven rotational transitions



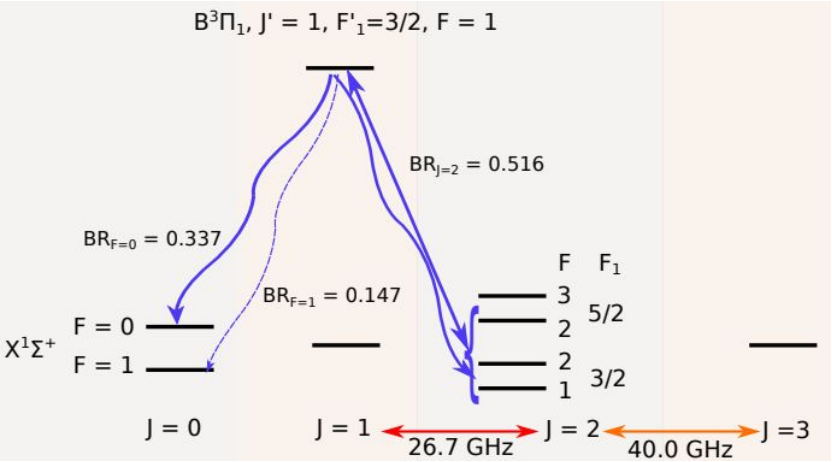


Fig. 6.2 of ref. **[4]**

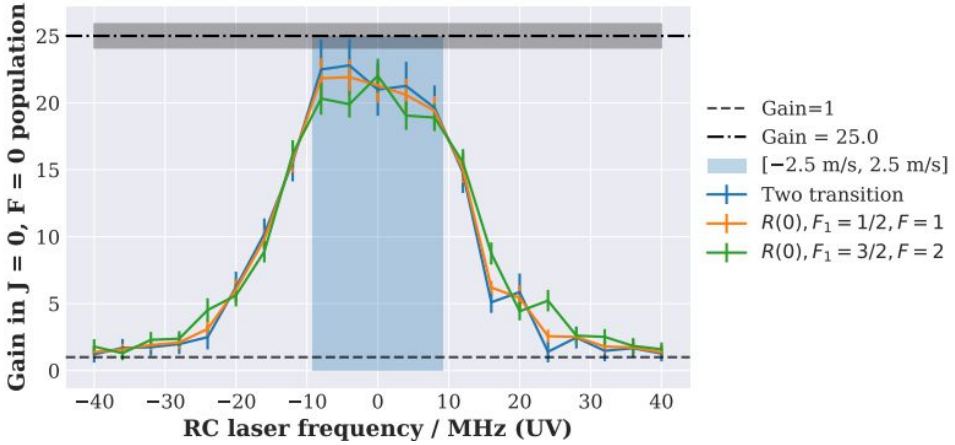


Fig. 6.12 of ref. **[4]**

SPA

Purpose: move population from $|J=0, F=0\rangle$ into $|J=2, m_J=0\rangle$, which is the state best focused by the lens.

Method: 2× adiabatic rapid passage

Step 1: $J=0$ to $J=1$

Step 2: $J=1$ to $J=2$

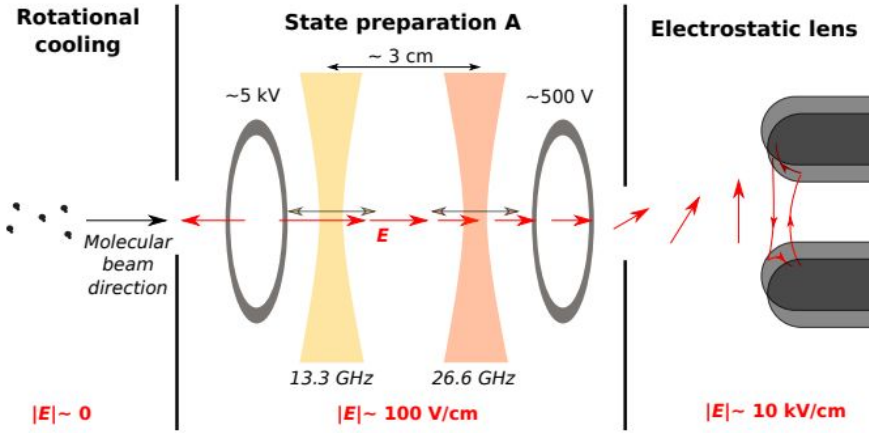


Fig. 7.14 of ref. **[4]**

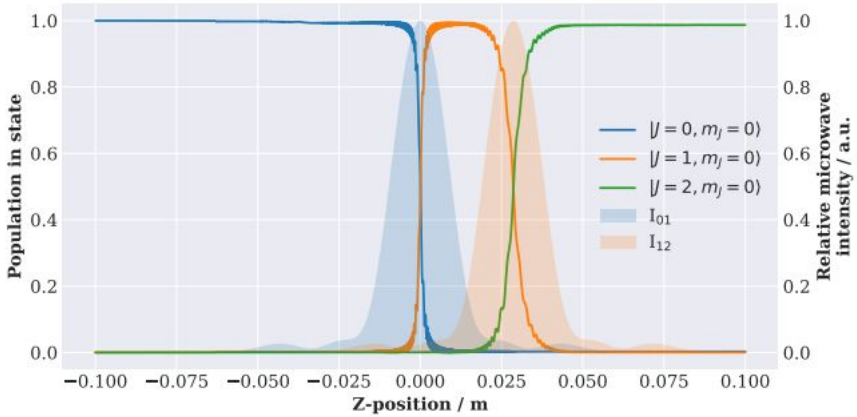


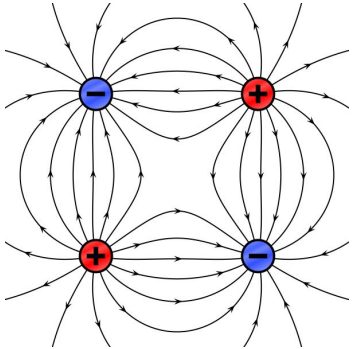
Fig. 7.19 of ref. **[4]**

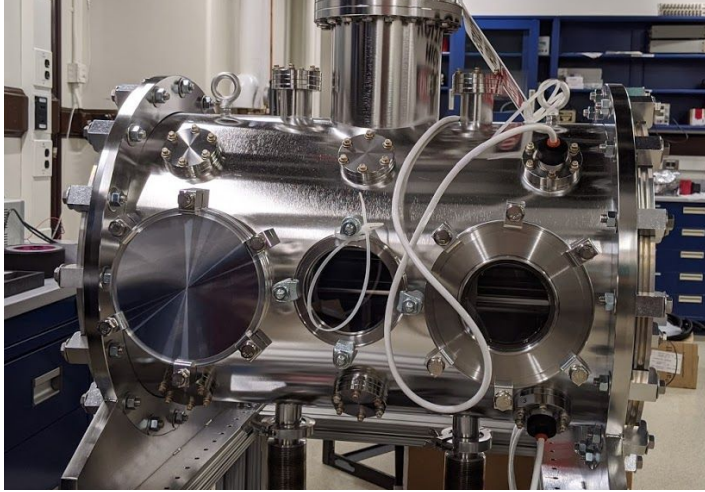
Lens

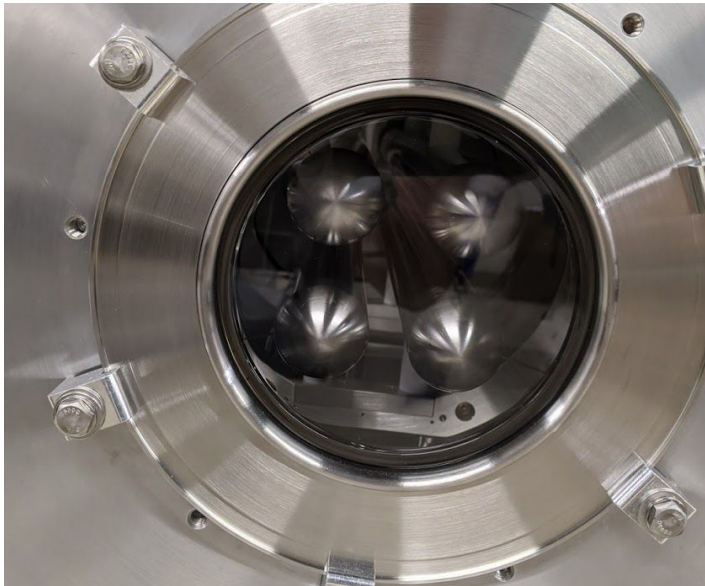
Purpose: focus the beam that moves through the electrostatic quadrupole field

Method: focusing based on quadratic Stark effect

Parameter	value	units
quadrupole voltage	±30	kV
bore diameter	1.75	in
length	60	cm







SPB

Purpose: move population from lens state $|J = 2\rangle$ into $|J = 1\rangle$ state needed for interaction region

Method: microwave π-pulse

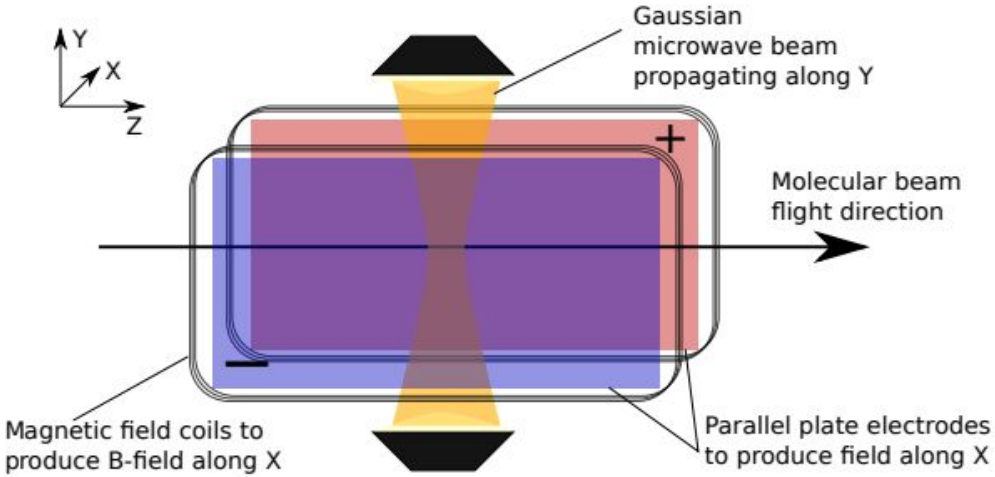


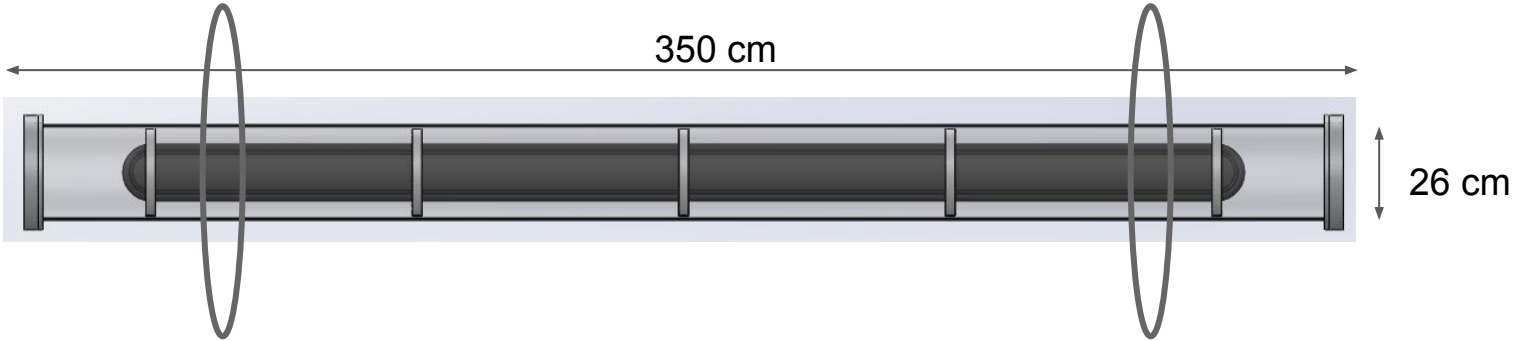
Fig. 9.2 of ref. **[1]**

Main interaction region

Purpose: induce hyperfine transitions sensitive to tiny energy differences

Method: Ramsey's separated oscillatory fields

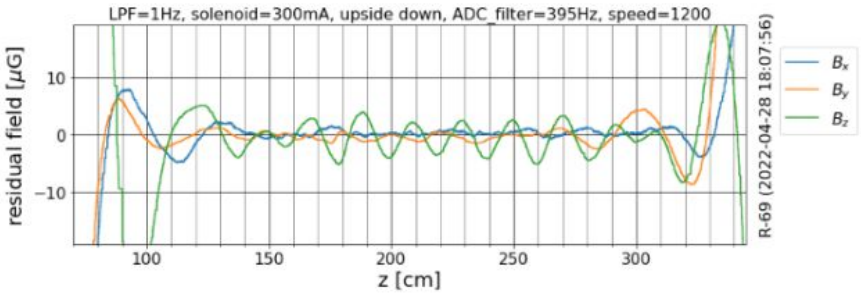
glass vacuum chamber:

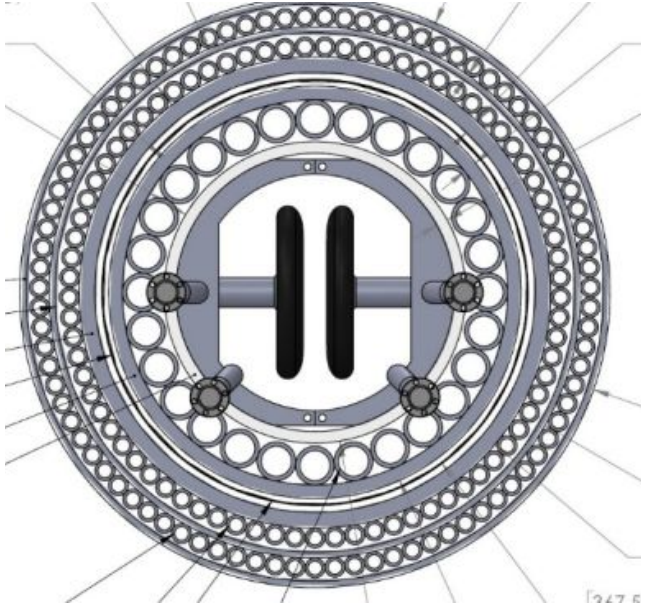


Magnetic shielding

Purpose: attenuate transverse lab magnetic fields down to 10 μG level

Method: 4-core cylindrical shields made from metglas and mu-metal





To be discussed in more detail shortly …

SPC & Detection

Purpose: map the two interaction-region states to different rotational levels, and then detect how much population was transferred

Method: laser-induced fluorescence

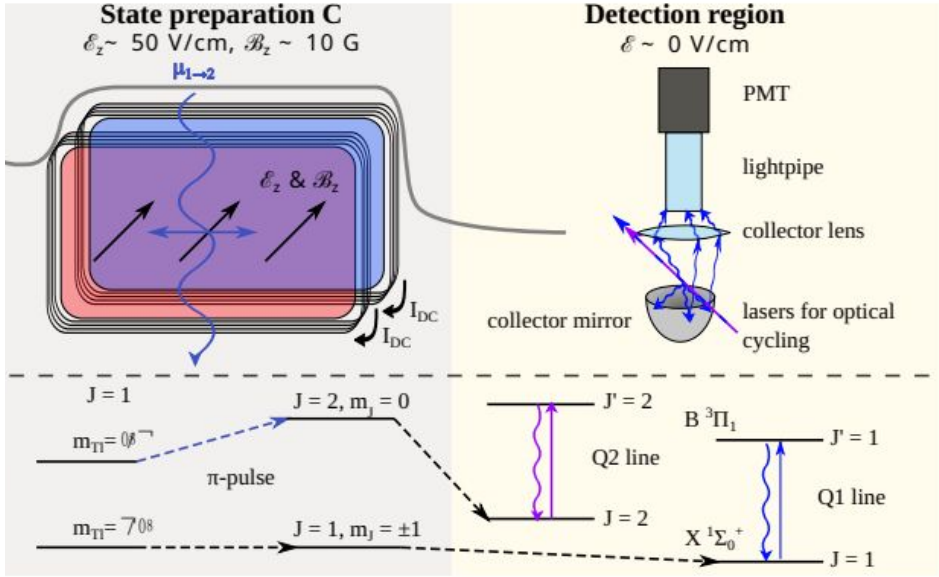


Fig. 6 of ref. **[1]**

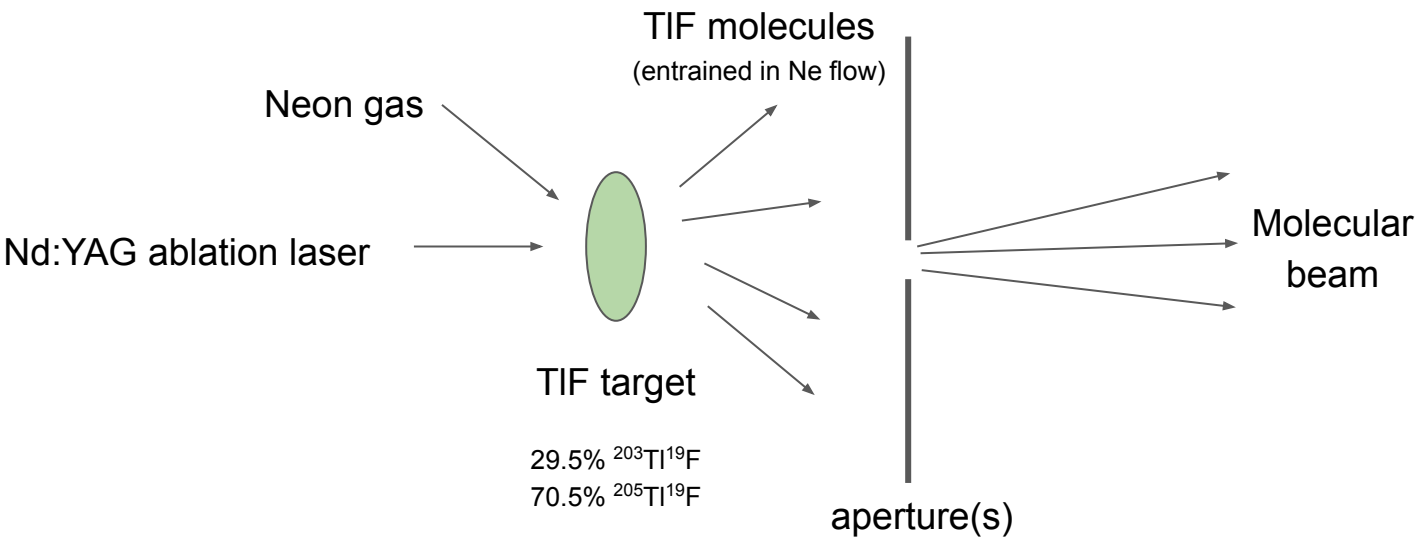
- Purpose of the experiment
- Experimental apparatus
- ➢ **Beam source**
- Magnetic shield
- Other topics
- Conclusion

Beam source

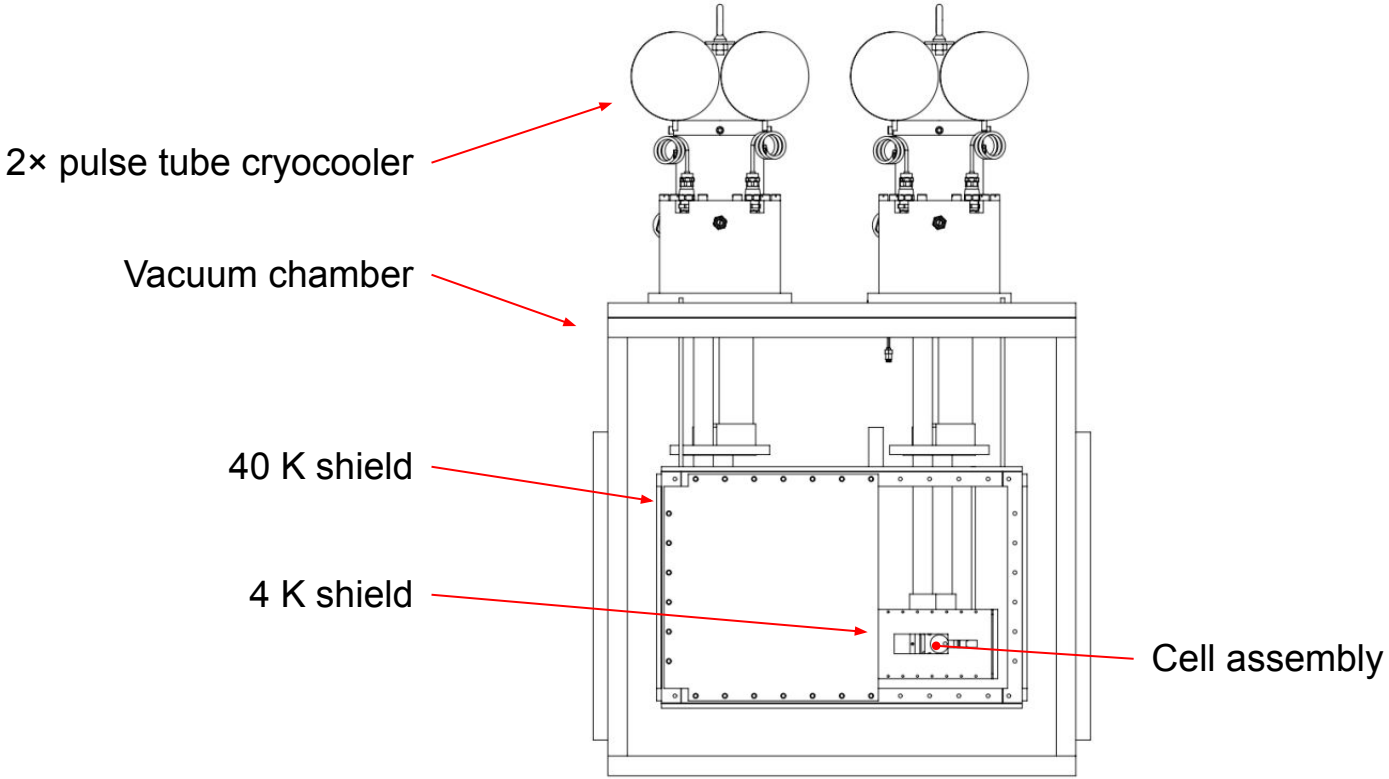
- Purpose
- Structure
- Target making
- Thermal
- Performance

Purpose

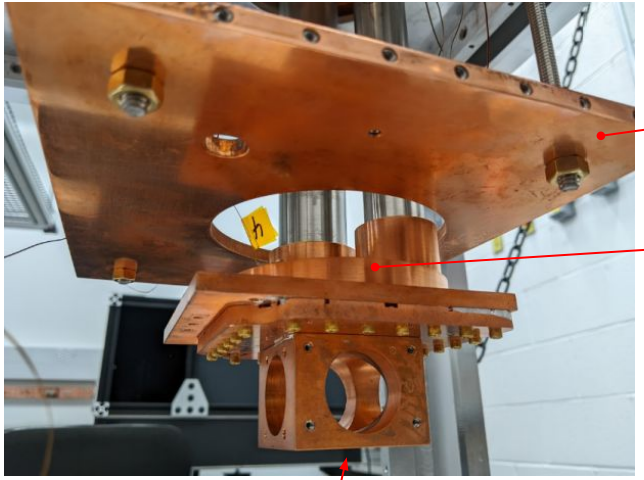
To create a cold molecular beam using cryogenic laser ablation of solid TlF.



Structure


2× pulse tube cryocooler
Vacuum chamber
40 K shield
4 K shield
Cell assembly

Cell assembly



Top plate of 4 K shield

Cold head of “16 K” pulse tube

Cell

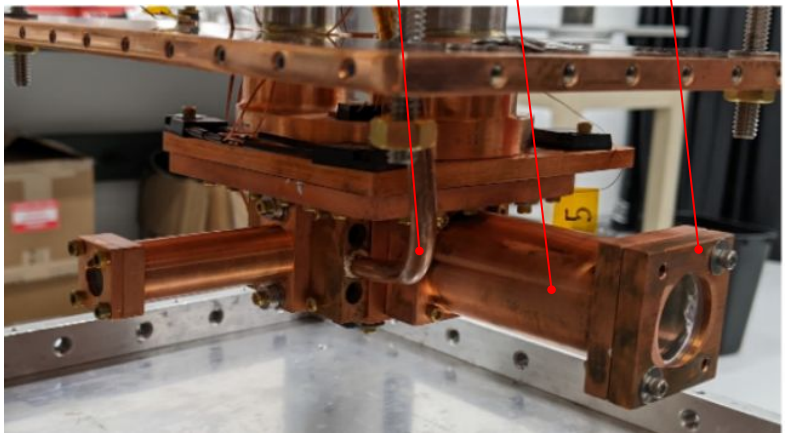
2” × 2” × 1.5” tall

ID = 1.15” circ. openings

Ablation laser window

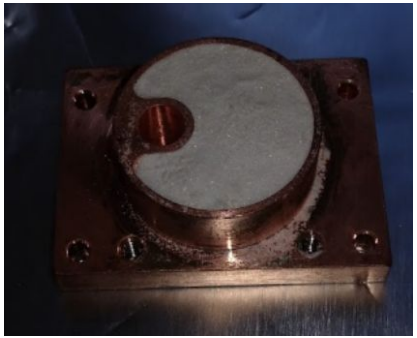
Gas line

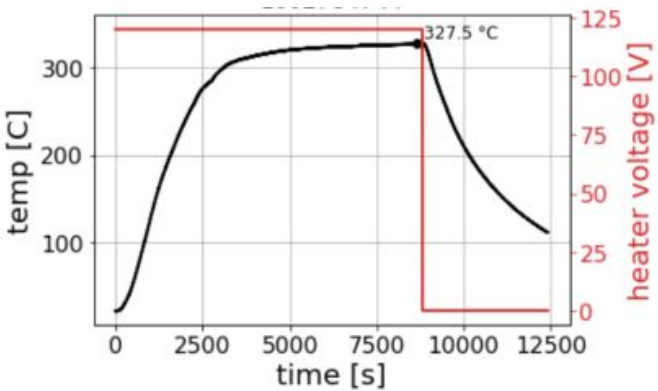
“Snorkel”

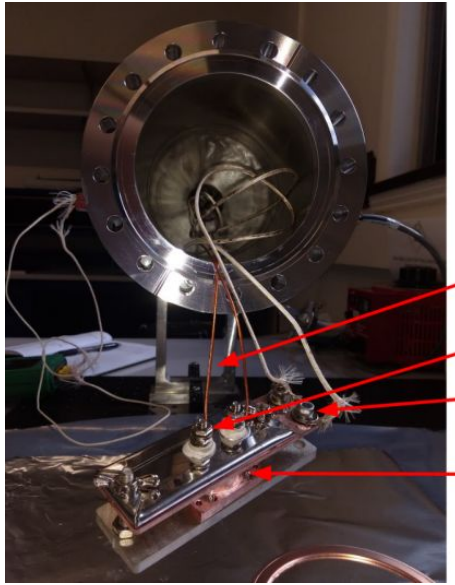


Target making

TIF powder before melting







kapton-insulated wire

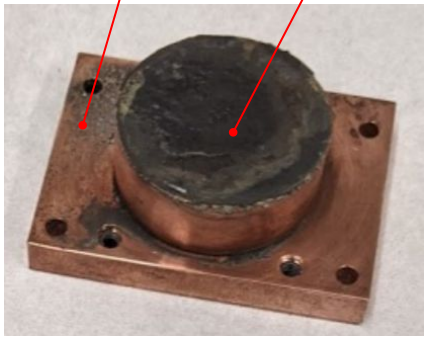
heater power connections

thermocouples

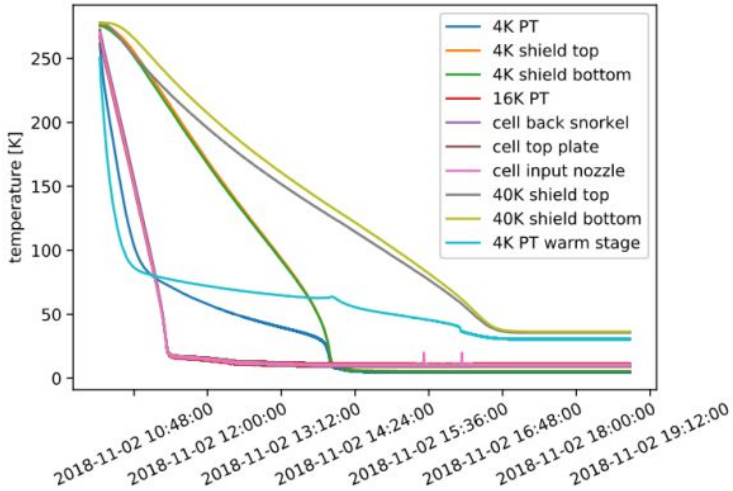
target inserted between the stainless and copper plate with the heater on top

Cu target holder

Solid TIF



Thermal


4K PT
4K shield top
4K shield bottom
16K PT
cell back snorkel
cell top plate
cell input nozzle
40K shield top
40K shield bottom
4K PT warm stage
temperature [K]
250
200
150
100
50
0
2018-11-02 10:48:00
2018-11-02 12:00:00
2018-11-02 13:12:00
2018-11-02 14:24:00
2018-11-02 15:36:00
2018-11-02 16:48:00
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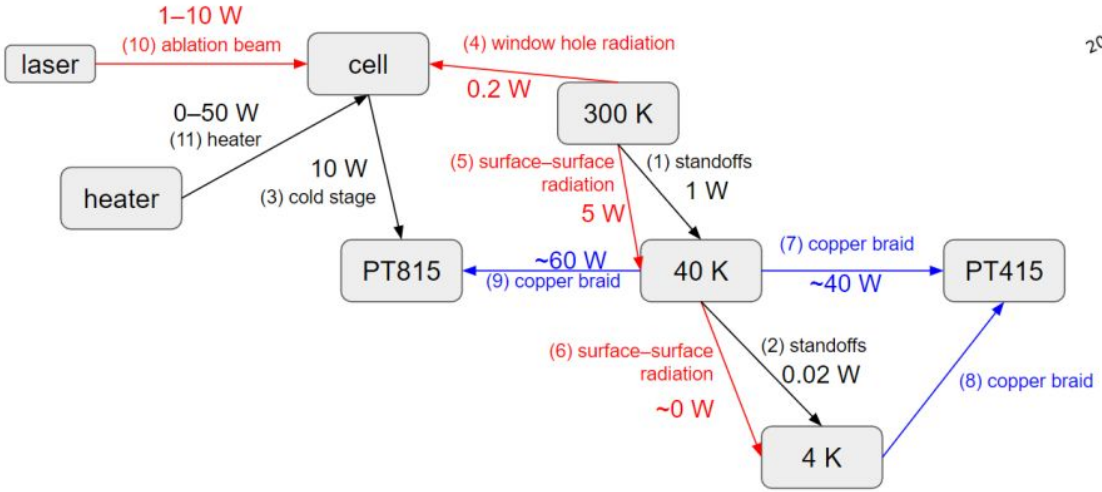
2018-11-02 19:12:00


laser
1–10 W
(10) ablation beam
cell
(4) window hole radiation
0.2 W
300 K
0–50 W
(11) heater
heater
10 W
(3) cold stage
(5) surface–surface radiation
5 W
(1) standoffs
1 W
PT815
~60 W
(9) copper braid
40 K
(7) copper braid
~40 W
PT415
(6) surface–surface radiation
~0 W
(2) standoffs
0.02 W
(8) copper braid
4 K

Performance

Parameter	value	units
Source intensity ($J = 0$ state)	$1.4 \cdot 10^{11}$	mol./sr/pulse
Average beam velocity	184 ± 17	m/s
Longitudinal velocity spread	16.1 ± 0.8	m/s
Transverse velocity spread	7.7	m/s
Translational temperature	7.0 ± 0.7	K
Rotational temperature	6.5 ± 0.2	K

Doppler shift measurements

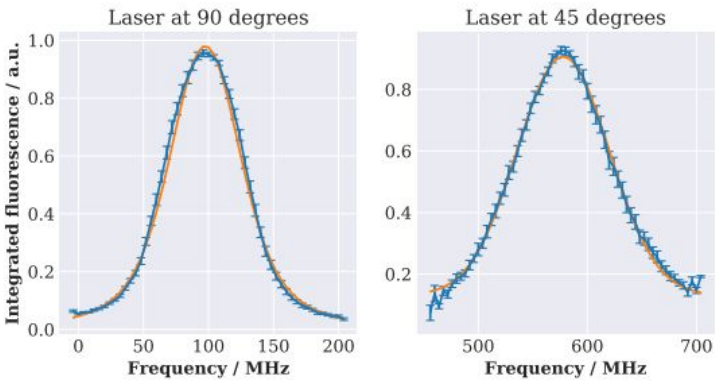


Fig. 5.5 of ref. **[4]**

Population vs rotational level

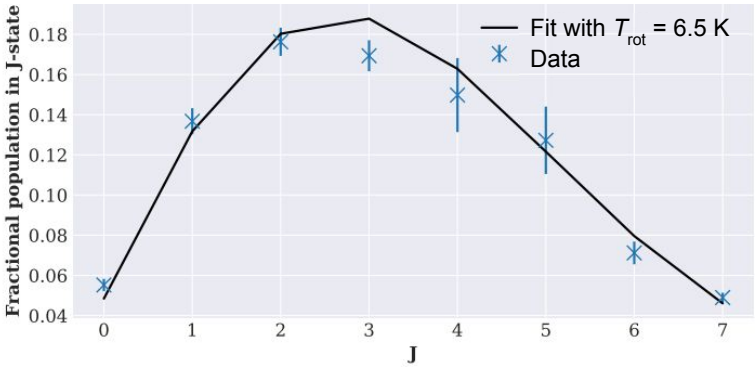


Fig. 5.9 of ref. **[4]**

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- ➢ **Magnetic shield**
- Other topics
- Conclusion

Magnetic shield

- Requirements
- Theory & degaussing
- Structure
- Performance
- Conclusion

Requirements (1): static field

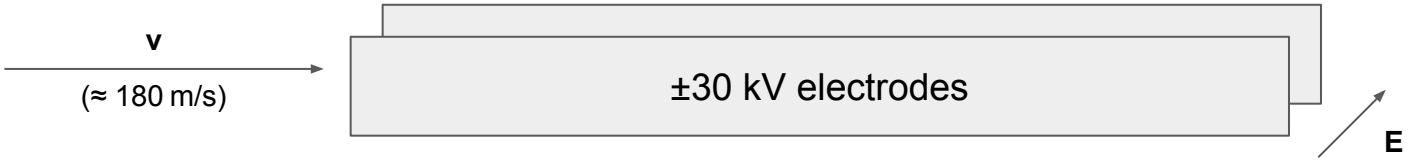
At velocity **v** in external field **E**, molecules experience **motional magnetic field:**

$$\mathbf{B}_{mot} = \mathbf{v} \times \mathbf{E} / c^2 \approx 60\ \mu G \qquad \text{(pointing up)}$$

which reverses direction with **E** and adds to lab field $\mathbf{B}_{lab}$, with magnitude depending on sign of **E**:

$$|\mathbf{B}_{total}| = |\mathbf{B}_{lab} \pm \mathbf{B}_{mot}|.$$

This can cause unwanted frequency shifts. → Want $|\mathbf{B}_{lab}| < 10\ \mu G$.



Requirements (2): noise

Sources of magnetic noise:

- Noisy lab magnetic fields
- Johnson noise in electric field plates
- Noise in magnetic field coils current

Estimate of allowed magnetic noise:

TIF magnetic moment ~ 1 kHz/G
desired experimental sensitivity ~ 50 nHz
∴ can allow up to **50 pG** of magnetic noise

This is noise integrated over a 1 μHz bandwidth (300 hours of data), so noise density < 50 nG/√Hz.

With typical lab noise ~ 10 μG/√Hz, need to attenuate noise by a factor SF ~ 200×.

Theory: linear materials

Inside a cylindrical shell, expect external transverse fields attenuated by a factor

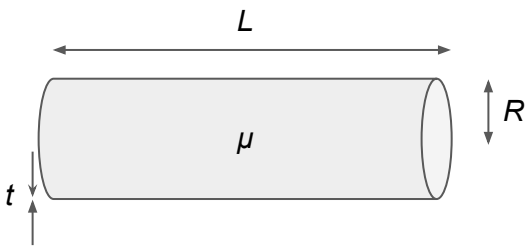
$$S_{\mathrm{T}} = \frac{\mu t}{2R}$$

With *n* concentric shells, the overall shielding is (almost) multiplicative:

$$S_{\mathrm{T}} = S_{\mathrm{T}}^{n} \prod_{i=1}^{n-1} S_{\mathrm{T}}^{i} \underbrace{\left[1 - \left(\frac{R_i}{R_{i+1}}\right)^2\right]}_{\text{“volume loss” in the region between the shells}}$$

Typical shield parameters (in CeNTREX)

Sym.	Name	Value	Unit
L	length	2–3	m
$2R$	diameter	14	in
t	thickness	0.2	mm
μ	permeability	70,000	—
S_{T}	shield factor	39.4	(theory)



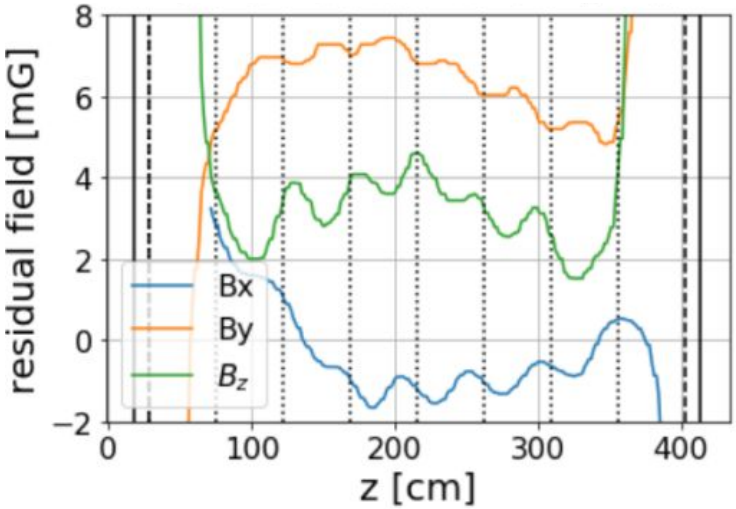
Example

In practice, lab field ~ 200 mG attenuated down to ~ 5 mG.

So SF ~ 50, just as predicted.

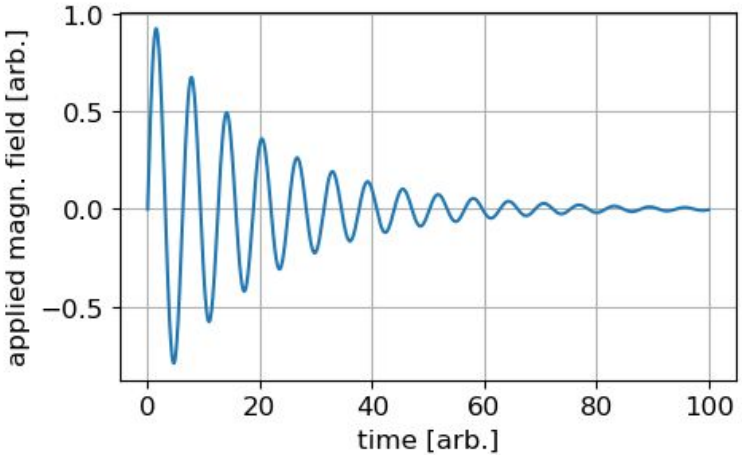
(Big error bars …)

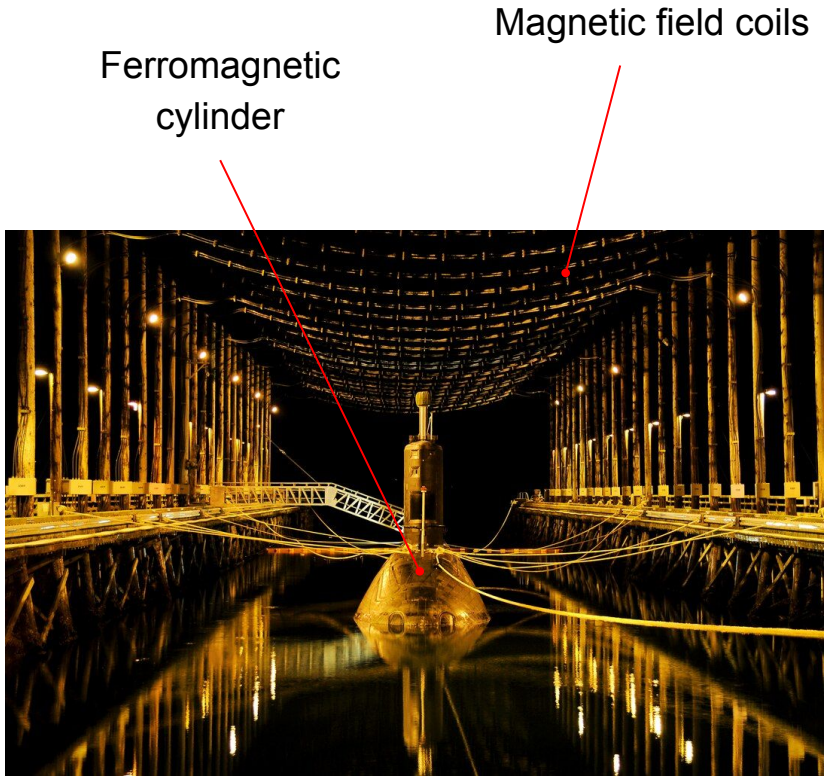
Can we do better than this?



Degaussing / deperming

Applying a decaying sinusoidal external magnetic field is known to “reduce the magnetization” of ferromagnetic objects.

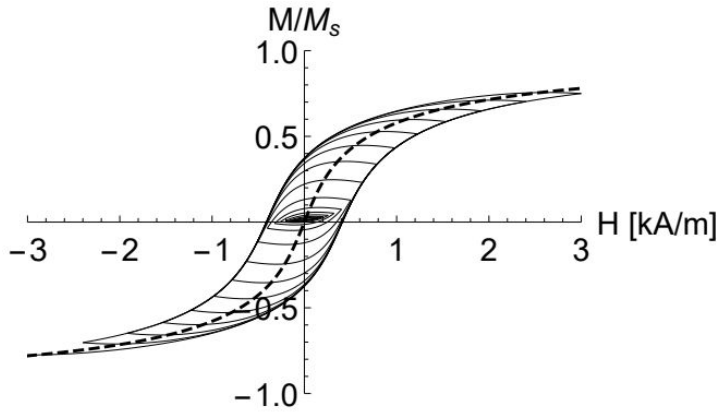




Credit: Department of Defense (2011) [https://dod.defense.gov/OIR/gallery/igphoto/2001186451]

How does it work?

The usual explanation
Magnetic material is driven around its hysteresis curve until it spirals into the origin. It is left with no magnetization, i.e. good shielding.

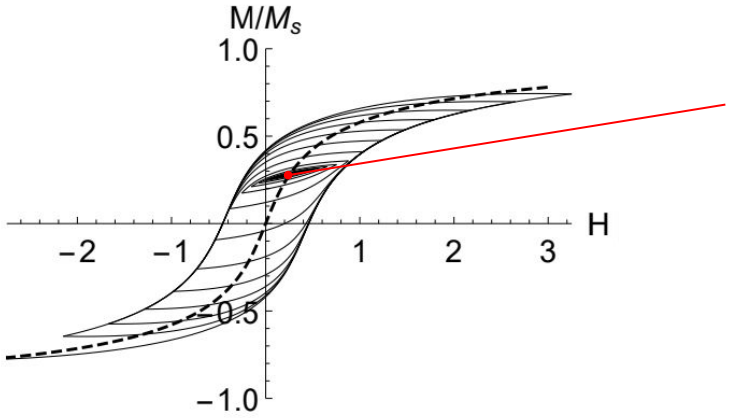


Two problems …

1. With **M** = 0, the shield has no magnetic effect on anything.
2. The applied field exponentially decays to a lab field ~ 500 mG, and not to zero.

More precisely …

Decaying sinusoid spirals the material onto a particular value magnetization that happens to oppose* the external magnetic field.



$\mathbf{M}_{final} \neq 0$ since $\mathbf{B}_{ext} \neq 0$.

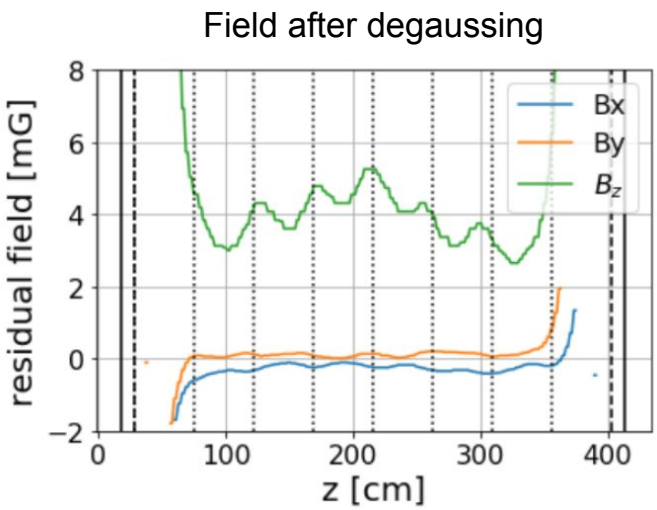
*Why? Because less magnetic field means lower total energy.

Surprise for linear theory!

Transverse components of the field ($\mathbf{B}_x$, $\mathbf{B}_y$) attenuated to ~ 0.2 mG.

With lab fields ~ 200 mG, this is a shielding factor of SF ~ 2000!

(Linear theory predicted SF ~ 40.)



What permeability?

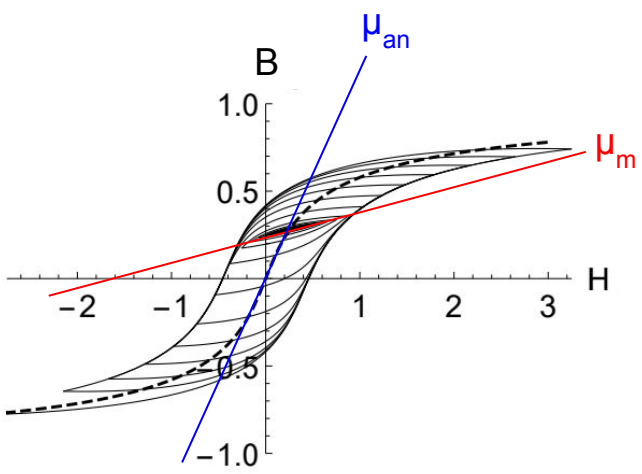
The shielding factor is proportional to the permeability μ:

$$S_{\mathrm{T}} = \frac{\mu t}{2R}$$

Permeability is the slope of the hysteresis curve:

$$\mu = \frac{d\mathbf{B}}{d\mathbf{H}}$$

But which slope do we take?



- Shielding with respect to small field perturbations → red line, μ_m.
- Static shielding after degaussing → blue line, μ_{an}.

Structure of CeNTREX shield

Four passive layers:

Metglas 1

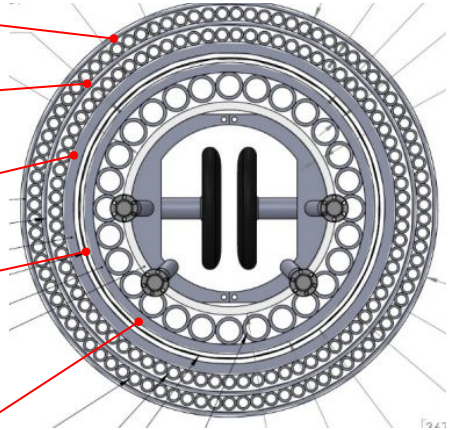
Metglas 2

Metglas 3

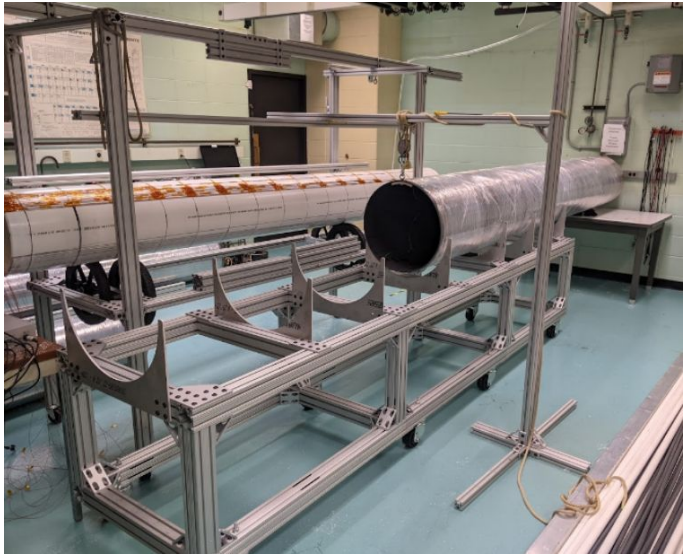
Mu-metal

One active layer:

Shim coils



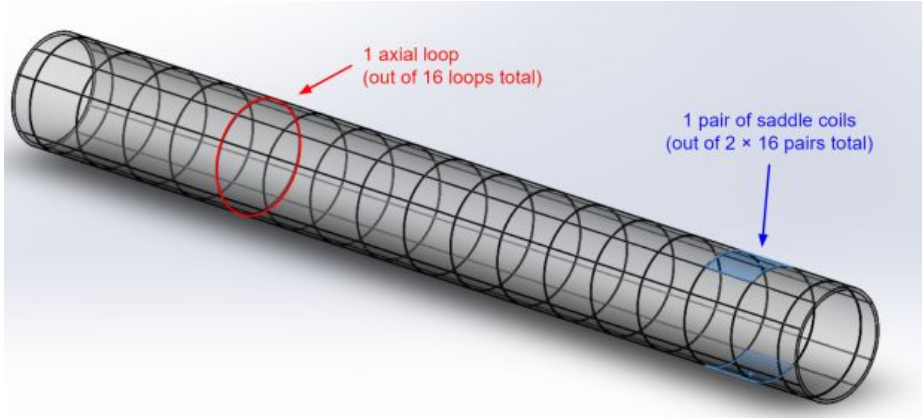
Tubes during assembly



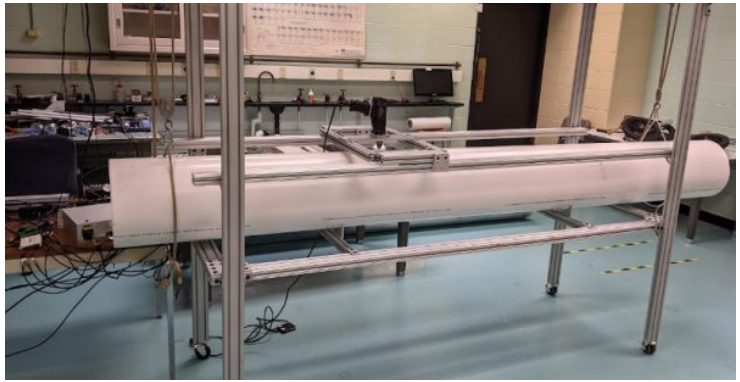
Shim coils

System of 5×16 single-loop coils:

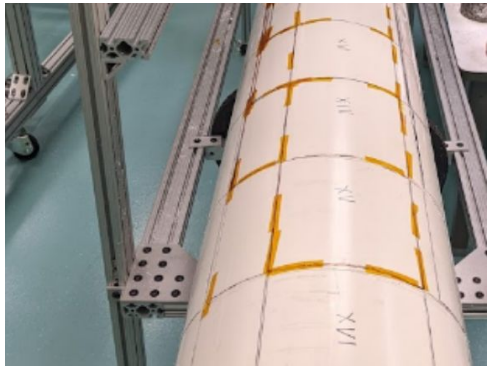
- 16 axial segments
- 2×2×16 saddle coil pairs



Grooves cut into the surface of PVC pipe

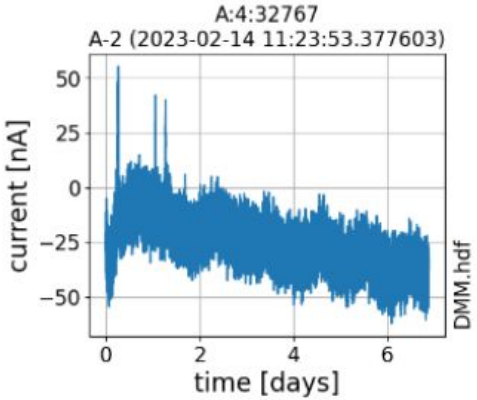


Wires glued into the grooves:

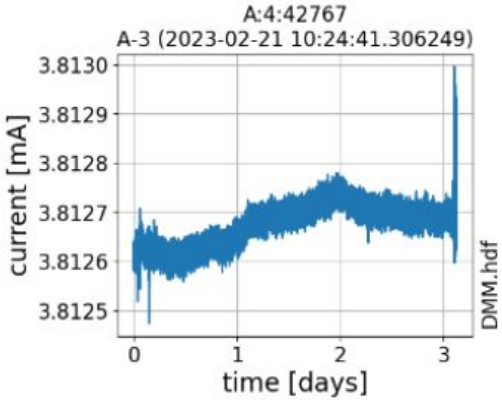


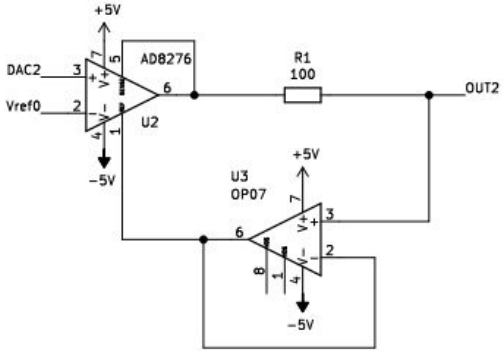
Shim coils current source

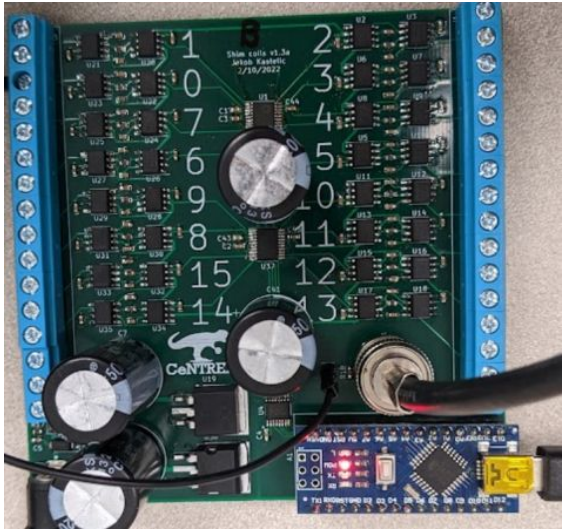
Mid-scale (≈ 0 mA)



⅓-scale (≈ 3.8 mA)





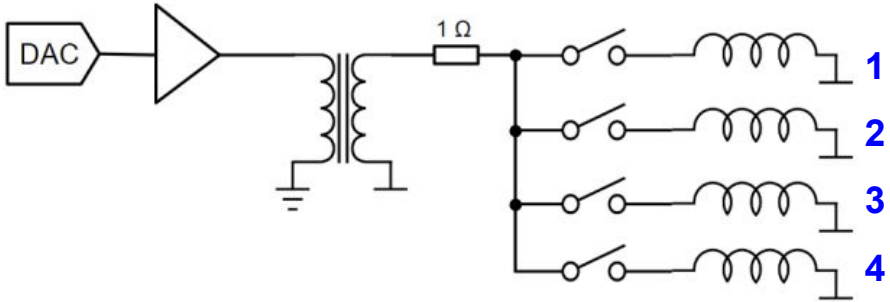


NB: shim coils response ~ 40 mG/A.

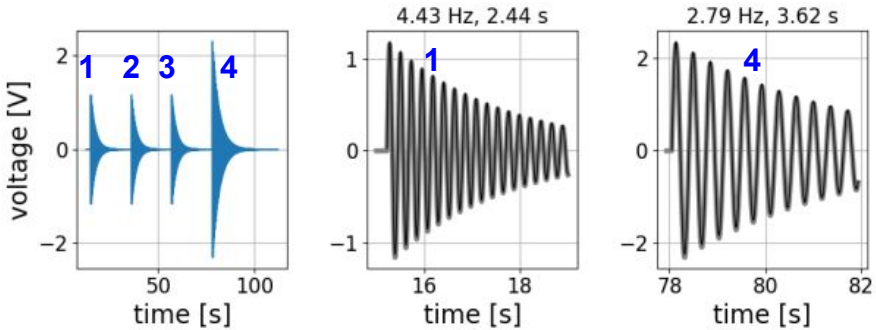
For < 0.1 μG field stability, need 2.5 μA current stability.

Degaussing coils & sequence

Toroidally wrapped around each cylinder:

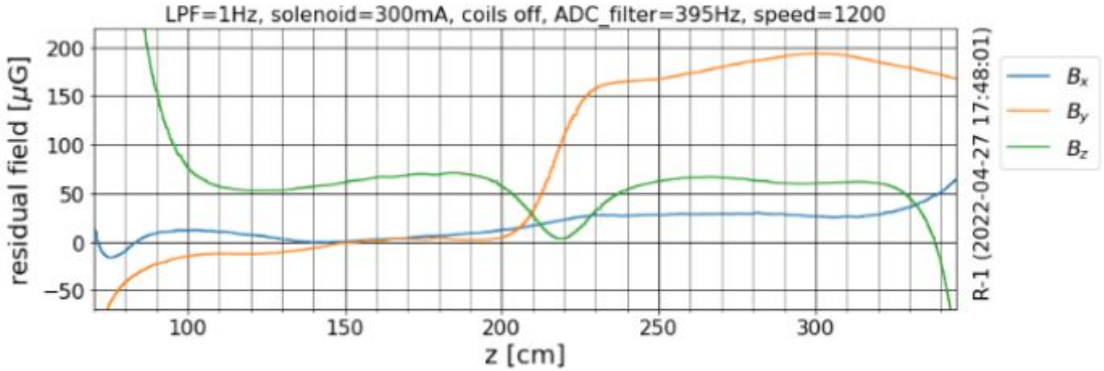


- 12 turns on each layer
- Degauss in sequence from outer to inner layer

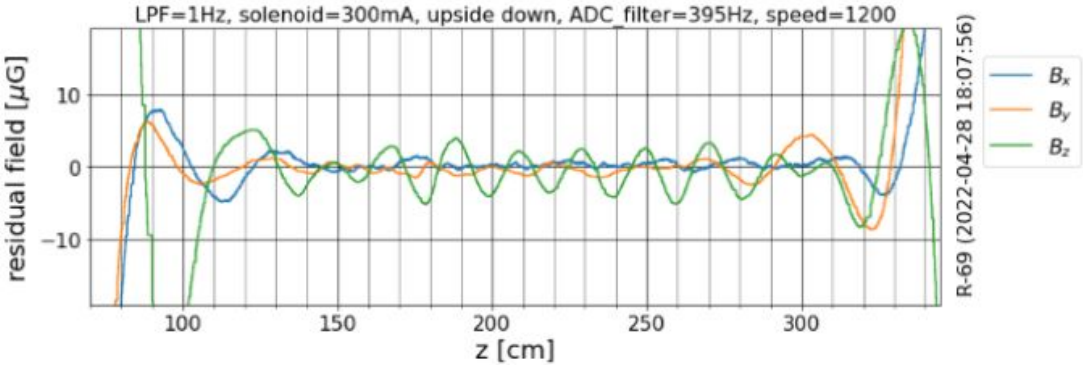


Performance 1: static residual field

Residual field after degaussing, unshimmed

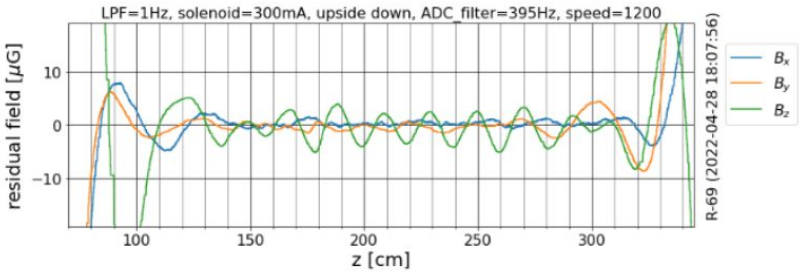


Shimmed

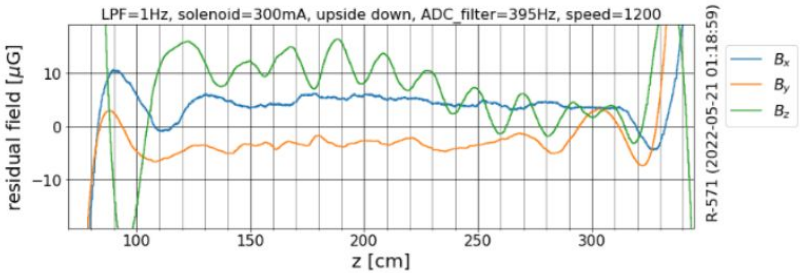


Performance 2: long-term stability

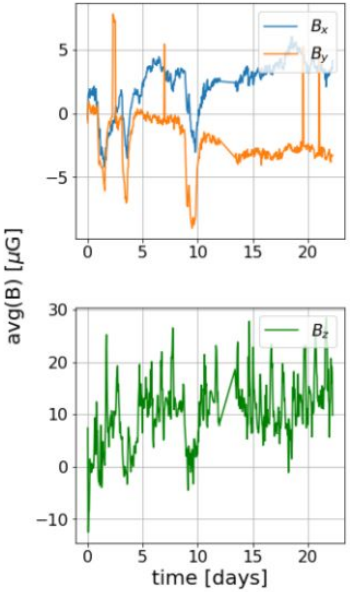
Initial



After 3 wks

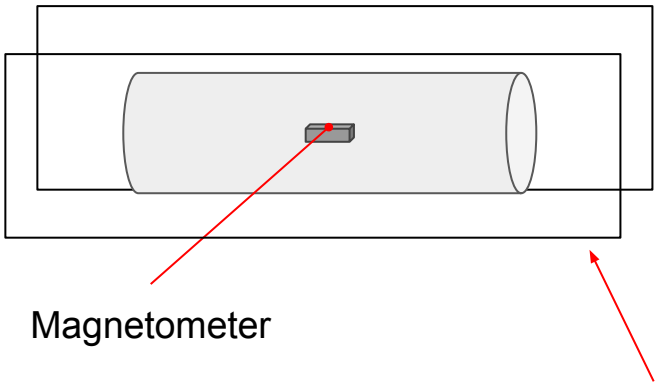


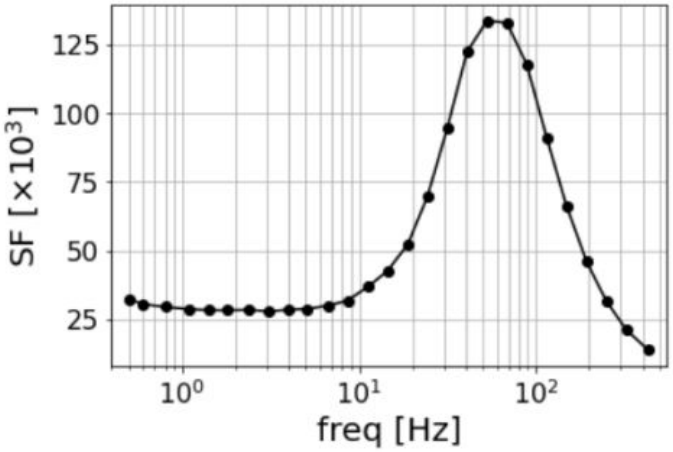
Average over central 2.5 m



Performance 3: dynamic shielding

$$\text{SF (dyn.)} = \frac{\text{applied field}}{\text{field measured inside}}$$





Conclusion

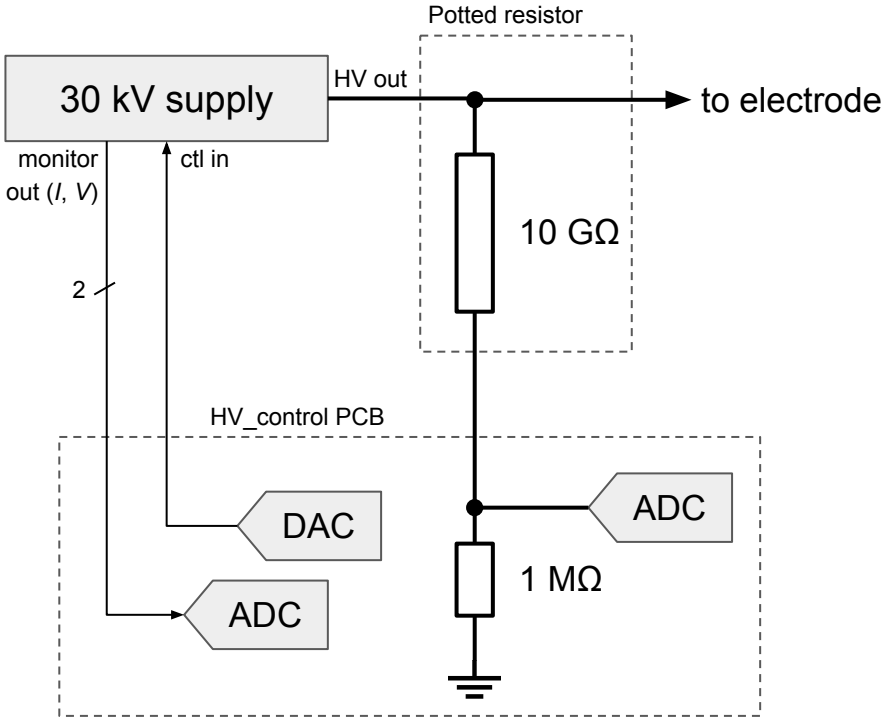
- Designed & built 4-layer magnetic shield
- Shimmed, the residual field achieves the < 10 μG target
- Field seems stable over the course of weeks
- Need more testing if long-term stability is reproducible

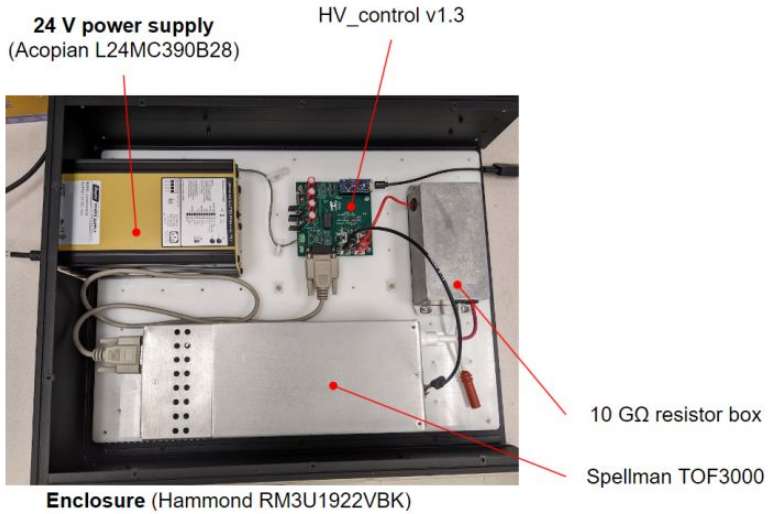
- Purpose of the experiment
- Experimental apparatus
- Beam source
- Magnetic shield
- ➢ **Other topics**
- Conclusion

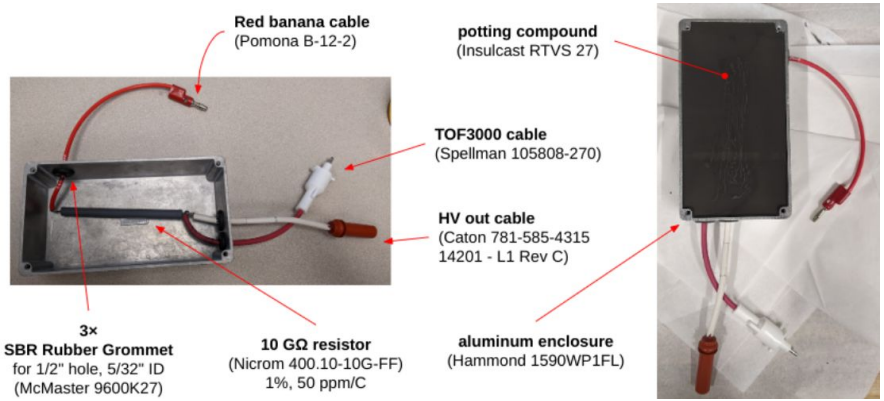
7. Other topics

- High voltage for lens
- Lens at HV
- Test interaction region
- PMT characterization

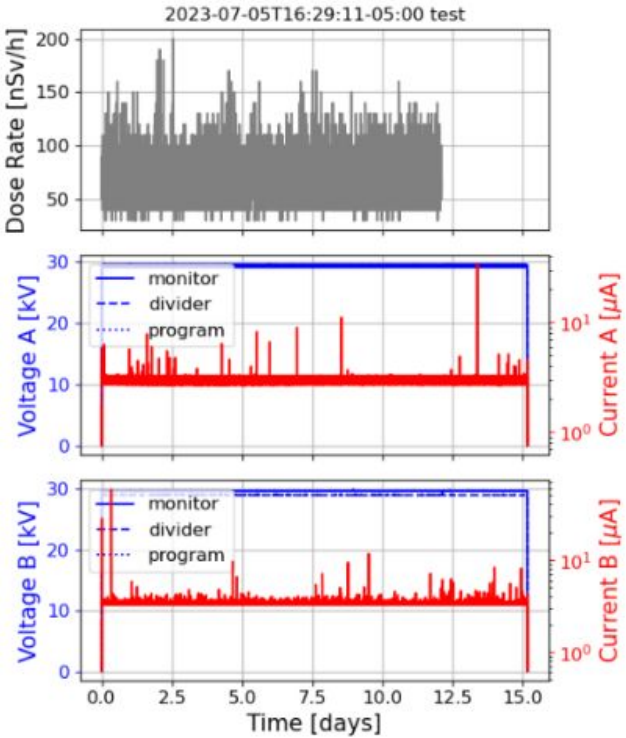
High voltage for lens







Lens at HV

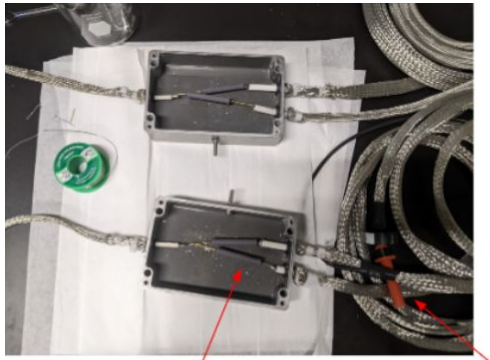


X-ray monitor

Thermo Scientific RadEye G20-10
gamma/X ray detector

Lesker VPZL-100 viewport
kodial window, 0.12" thick,
2.5" diam.

HV “splitter” boxes

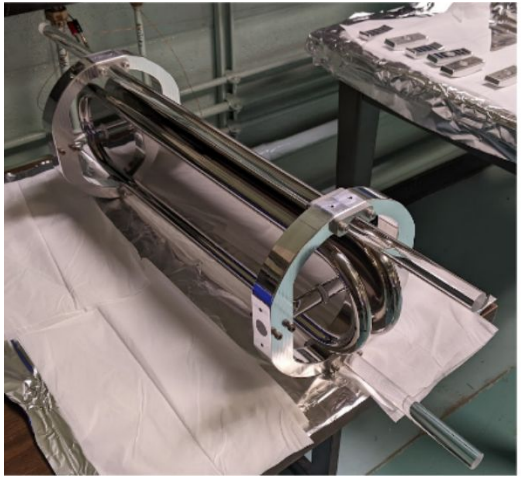


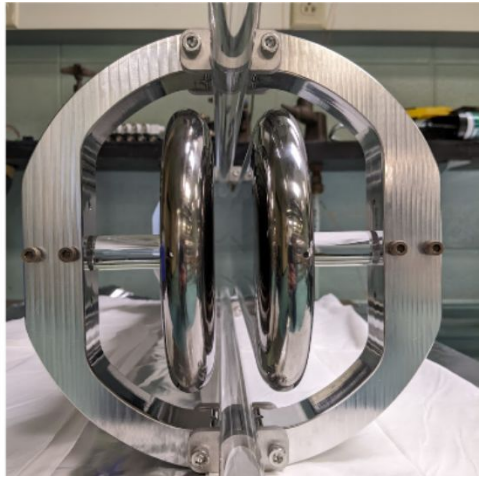
Nicrom 400.5-200M-FS
200M 30kV 7.5W 100 ppm/C

Ceramtec 21112-01-A
50 kV Cable Assembly - 14 AWG

Test interaction region

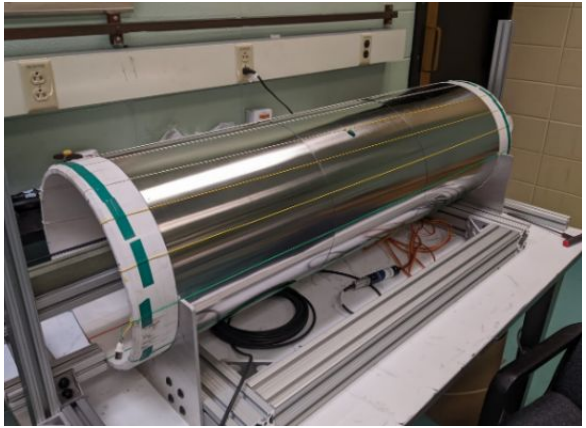
Purpose: enable testing the entire experiment without having the full-scale (3.5 m long) main interaction chamber ready.

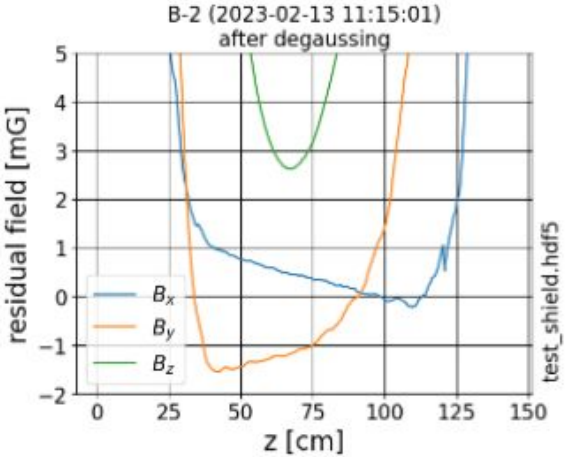




Electrode assembly (≈60 cm long, titanium)

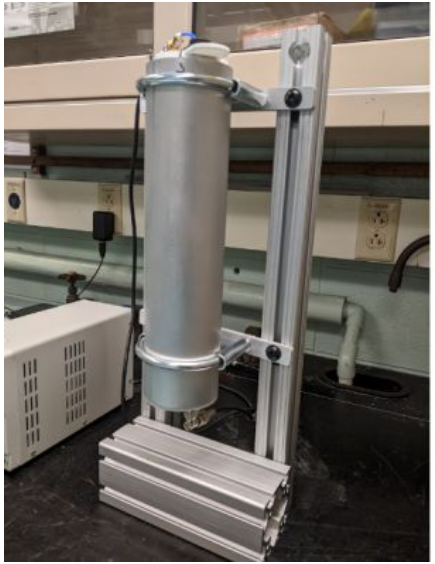
1-layer magnetic shield



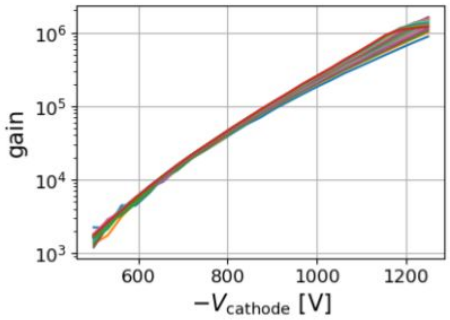


PMT characterization

- Use five PMTs in detection
- Want each PMT to have the same gain
- Measured the response vs V_{cathode} for each PMT



$$\text{gain} = \frac{a \cdot (-V_{\text{cathode}}/100\ \text{V})^k}{\int \rho_{\text{LED}} \cdot R_{\text{PMT}}\, \mathrm{d}\lambda}$$



$$I_{\text{anode}} = a \cdot P_{\text{light}}\,(-V_{\text{cathode}}/100\ \text{V})^k$$

Serial	a [mA/W]	k	V_{cathode}^{*} [V]	gain [10^5]
SB4713	?	?	?	?
SB4714	0.652	6.838	1124.2	0.9
SB4715	1.086	7.224	920.5	3.7
SB4759	0.933	7.018	1004.0	2.0
SB4761	0.614	7.186	1008.3	1.9
SB4762	0.629	7.174	1008.8	1.9

* For 10 μA of anode current @ 1 nW light input.

- Purpose of the experiment
- Experimental apparatus
- Beam source
- Magnetic shield
- Other topics
- ➢ **Conclusion**

Conclusion

- CeNTREX experiment is a new search for *T*-violating phenomena using cold molecules
- Schiff moment quantifies the “amount” of *T*-violating effects observable in a neutral atom/molecule
- CeNTREX apparatus is 75% built, still need to finish testing SPC/detection before starting to take data.
- Magnetic shield shown to attenuate lab fields down to the 10 μG level needed for CeNTREX

References

[1] Olivier Grasdijk, Oskari Timgren, Jakob Kastelic, Trevor Wright, Steve Lamoreaux, David DeMille, Konrad Wenz, Michael Aitken, Tanya Zelevinsky, Tristan Winick, et al. Centrex: a new search for time-reversal symmetry violation in the 205Tl nucleus. Quantum Science and Technology, 6(4):044007, 2021. doi:10.1088/2058-9565/abdca3.

[2] Schwartz, M.D. (2014). "29". Quantum Field Theory and the Standard Model.

[3] VA Dzuba, VV Flambaum, JSM Ginges, and MG Kozlov. Electric dipole moments of Hg, Xe, Rn, Ra, Pu, and TlF induced by the nuclear schiff moment and limits on time-reversal violating interactions. Physical Review A, 66(1):012111, 2002. doi:10.1103/PhysRevA.66.012111.

[4] Oskari Timgren. Progress towards a measurement of time-reversal symmetry violation in thallium fluoride. PhD thesis, Yale University, May 2023